

CUSTOMER SEGMENTATION USING K-MEANS AND HIERARCHICAL CLUSTERING

Course: MACHINE LEARNING – BTech
Academic Year: 2026–27

By:

B.Jhahnavi – Roll No:2520030452

P.Mouvya Sri– Roll No:2520030449

Guide: Shaik Asif

Department: Computer Science & Engineering

PROBLEM STATEMENT:

- Customer segmentation is an important task for businesses because customers have different purchasing behaviors, preferences, and spending patterns.
- The objective of this project is to develop a machine-learning-based customer segmentation system that groups customers with similar characteristics and purchasing behavior
- The system will help businesses understand different customer groups and develop targeted marketing strategies, personalized offers, and better customer relationship management.

PROPOSED SOLUTION

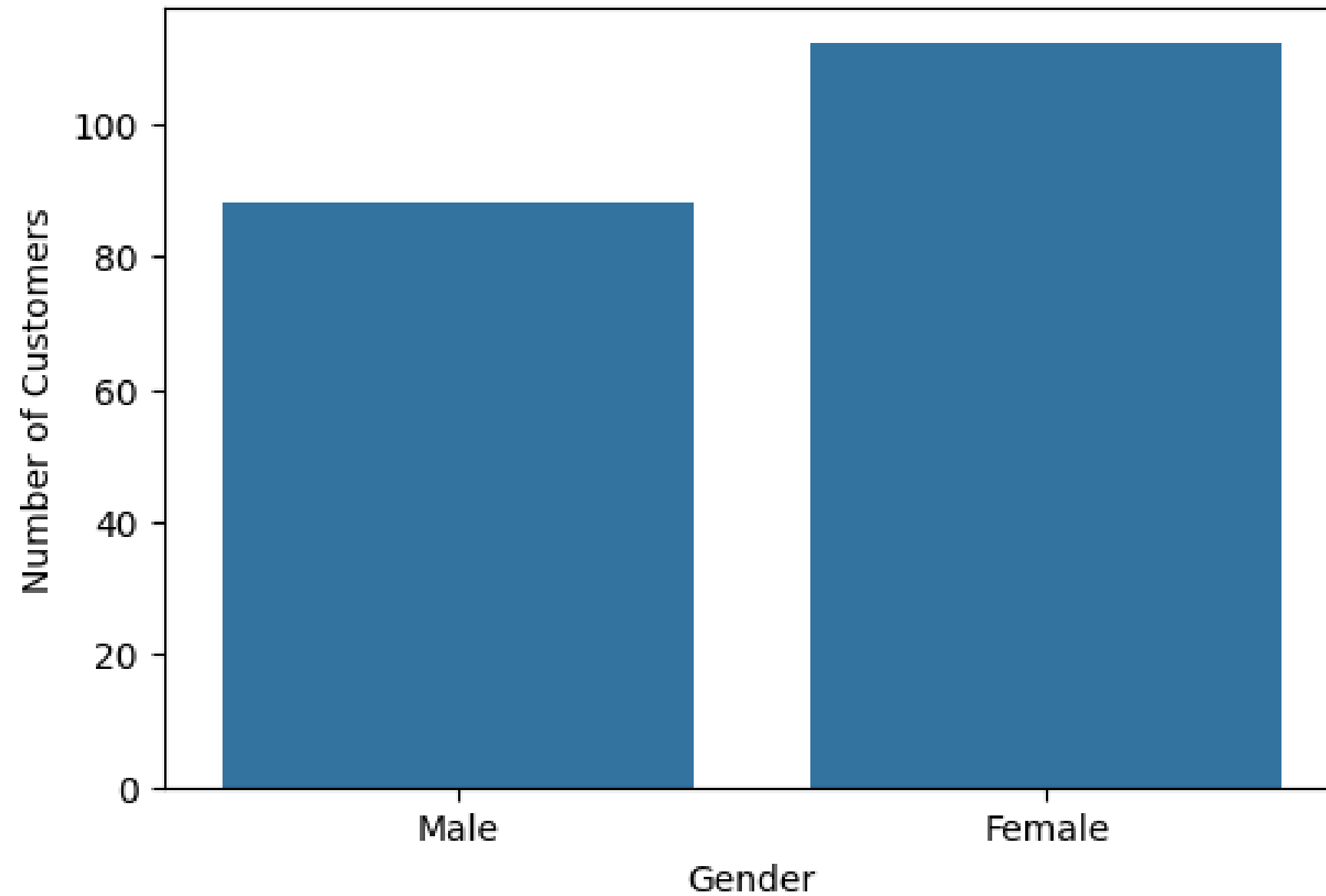
- We propose a machine-learning-based customer segmentation system using K-Means Clustering and Hierarchical Clustering.
- The system analyzes customer attributes such as:
 - Age
 - Annual Income
 - Spending Score
 - Purchase Frequency
 - Customer Behavior
- K-Means groups customers into a predefined number of clusters based on similarity.
- Hierarchical Clustering creates a tree-like structure of customer groups and helps visualize relationships between customers.
- The performance and cluster quality of both methods are analyzed and compared using suitable clustering evaluation metrics.

OBJECTIVES:

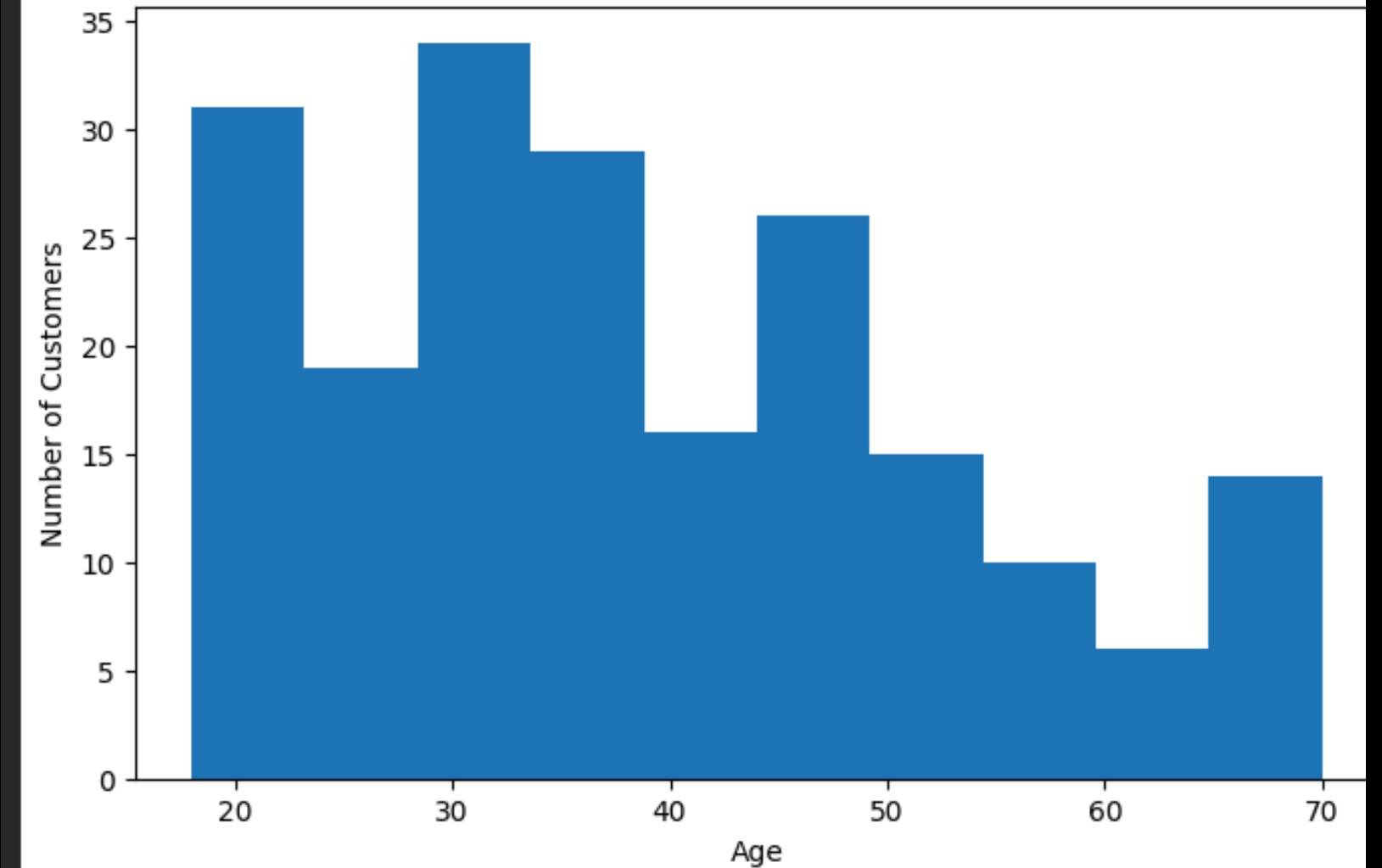
- To segment customers based on their characteristics and purchasing behavior.
- To analyze customer income and spending patterns.
- To apply K-Means Clustering for customer grouping.
- To apply Hierarchical Clustering for customer grouping.
- To determine the optimal number of customer segments.
- To compare K-Means and Hierarchical Clustering.
- To identify valuable, regular, and low-engagement customer groups.
- To help businesses develop targeted marketing strategies.

EXPLORATORY DATA ANALYSIS(EDA)

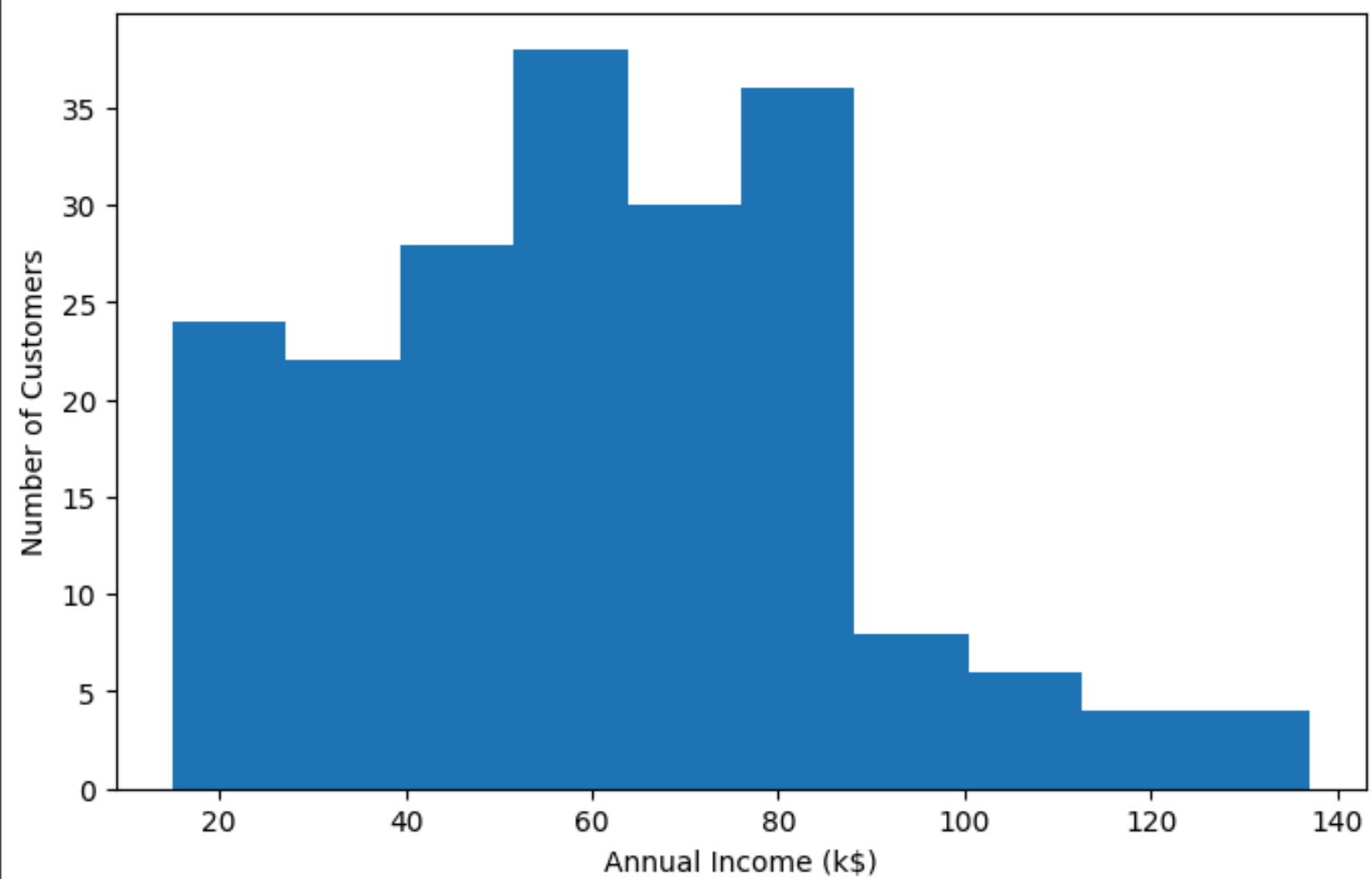
Gender Distribution



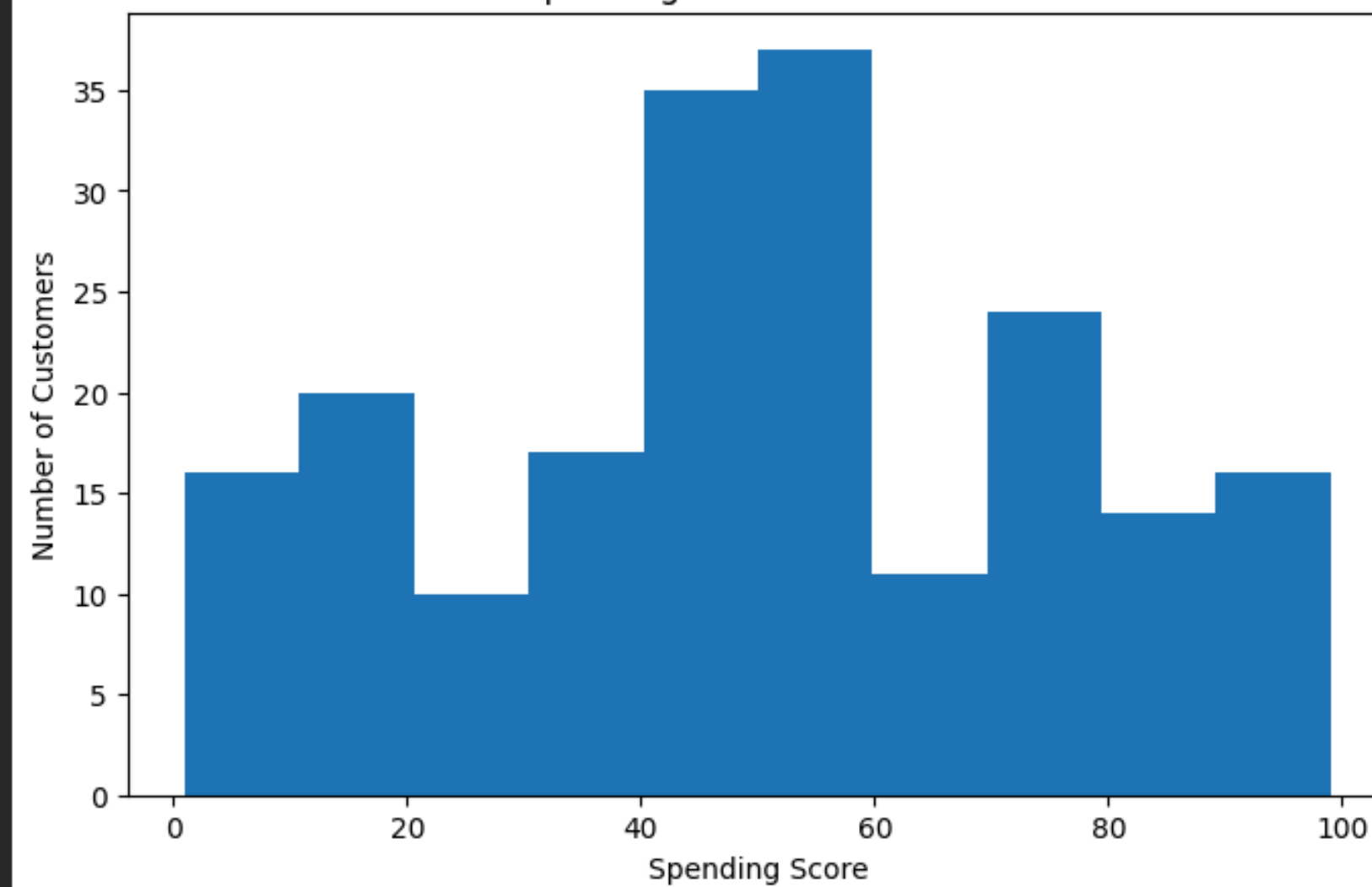
Age Distribution of Customers

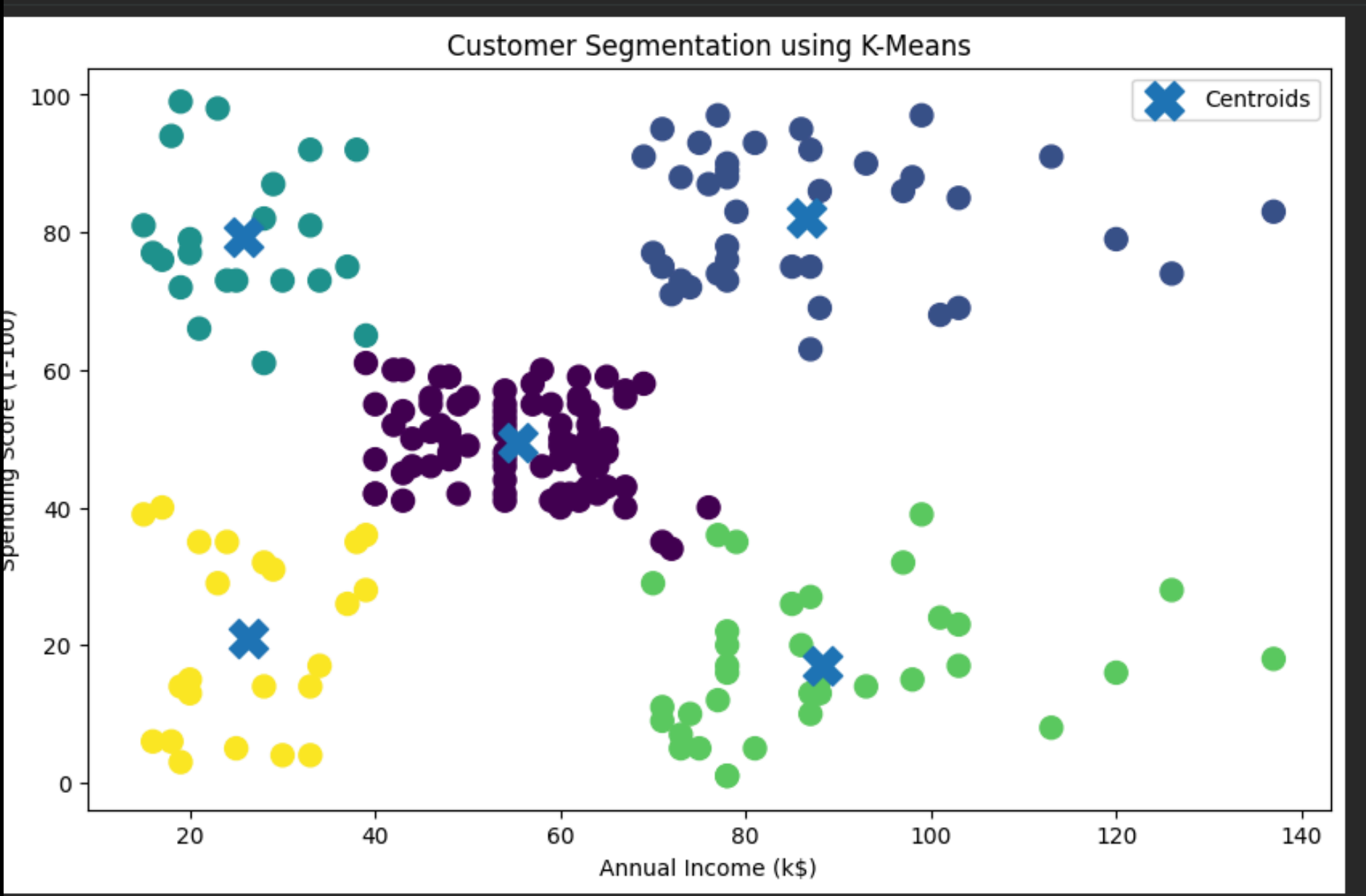
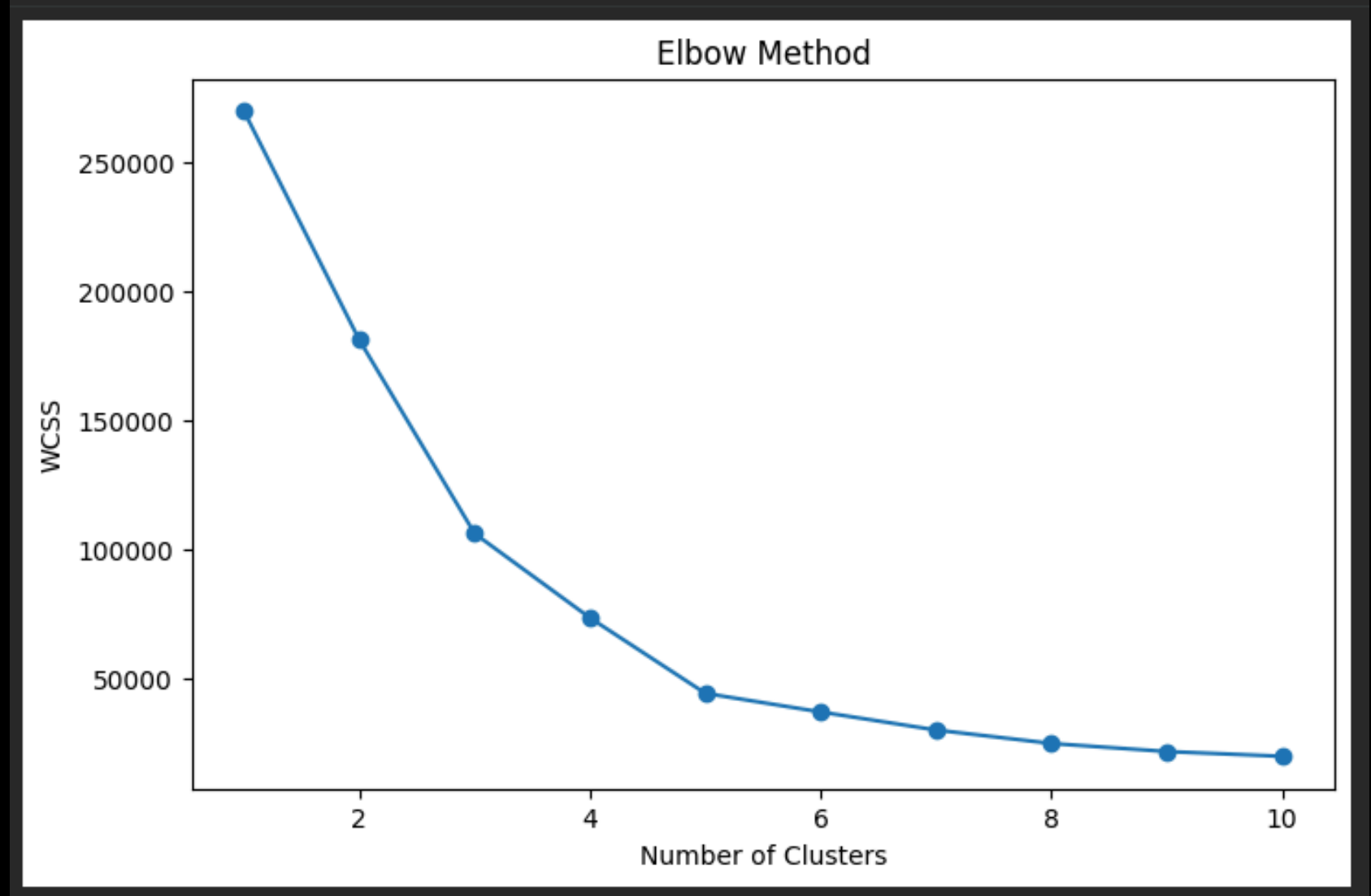


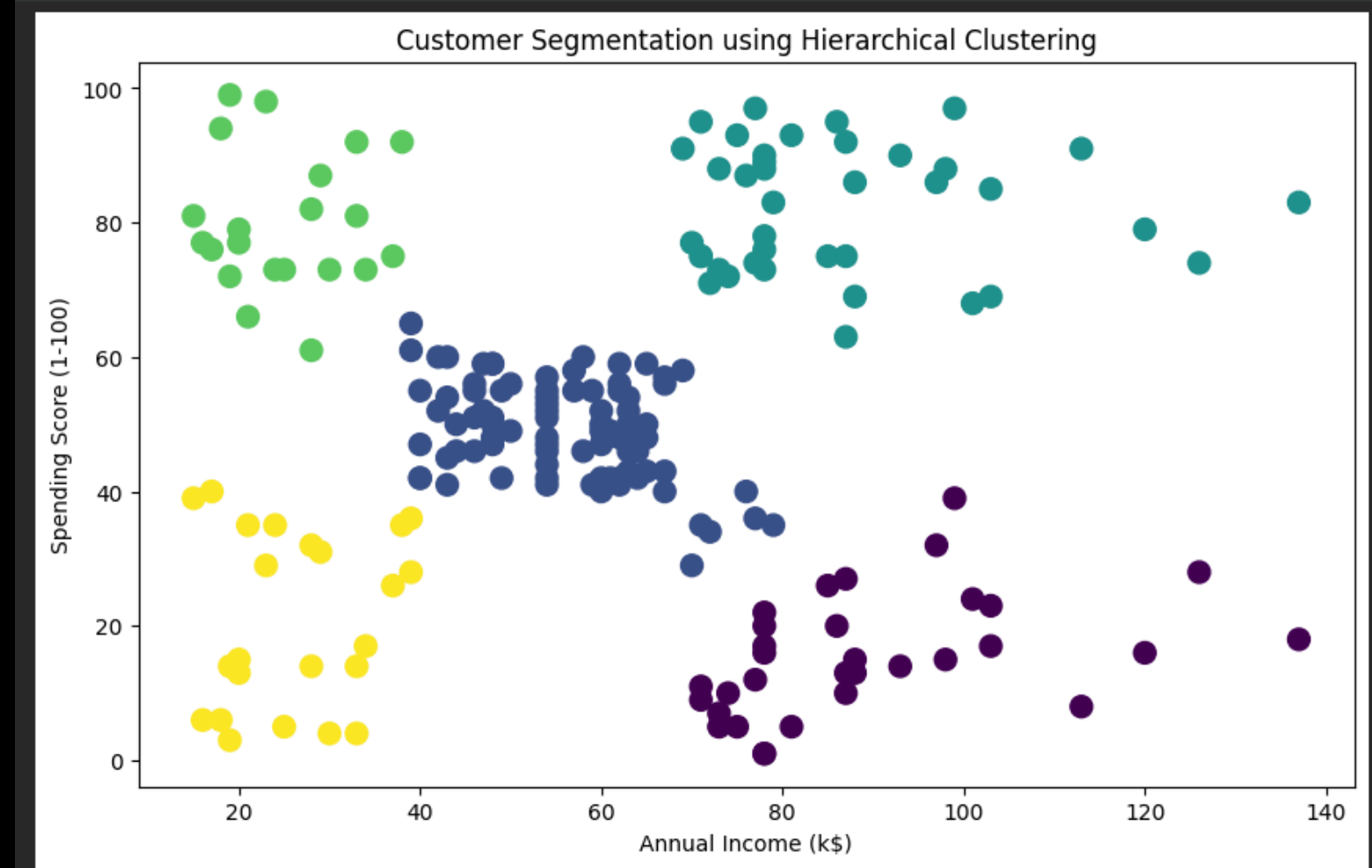
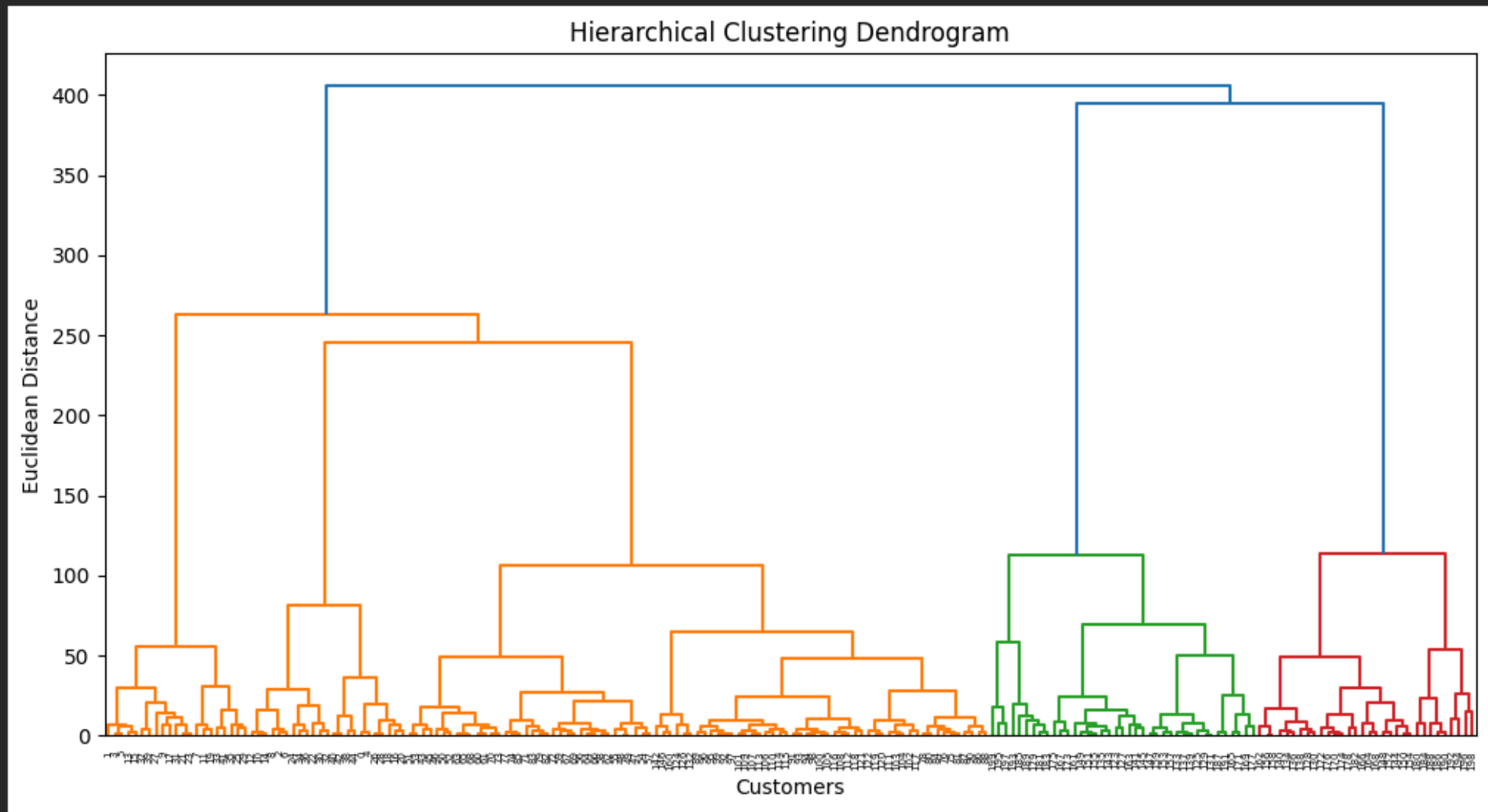
Annual Income Distribution



Spending Score Distribution







Modules, Objectives & Expected Outcomes

	Module	Objective	Expected Outcome
1	Module 1 — Data Acquisition & Exploration	- Collect and understand customer data. - Analyze data distributions and relationships. - Check for missing values and duplicates.	- Cleaned dataset. - Basic statistics and data summary. - Exploratory analysis with visualizations.
2	Module 2 — Data Preprocessing	- Encode categorical variables (e.g., Gender). - Select relevant features for clustering. - Scale numerical features.	- Prepared dataset suitable for clustering algorithms. - Transformed feature set. - Preprocessing pipeline.
3	Module 3 — K-Means Clustering	- Apply K-Means clustering. - Determine the optimal number of clusters using Elbow Method and Silhouette Score. - Analyze cluster characteristics.	- Customer segments generated using K-Means. - Elbow curve and silhouette score. - Cluster visualization and interpretation.
4	Module 4 — Hierarchical Clustering	- Apply Agglomerative Hierarchical Clustering. - Analyze customer relationships using dendrogram. - Form clusters and interpret results.	- Hierarchical customer segments. - Dendrogram visualization. - Cluster visualization and interpretation.
5	Module 5 — Evaluation & Comparison	- Compare K-Means and Hierarchical Clustering. - Evaluate clustering performance using silhouette score. - Interpret and profile customer segments.	- Comparison of both algorithms. - Final customer segments with insights. - Recommendations for business strategies.

RESEARCH PAPERS:

https://www.researchgate.net/publication/395121134_Mall_Customer_Segmentation_Using_K-Means_Clustering_Optimized_by_the_Elbow_Method?utm_source=chatgpt.com

https://www.preprints.org/manuscript/202604.1238?utm_source

EDA FINDINGS:

- 200 customers with 5 features.
- No missing values or duplicate records.
- 112 females and 88 males.
- Annual income ranges from \$15k–\$137k.
- Spending scores range from 1–99.
- Average annual income is \$60.56k and average spending score is 50.20.
- Age has a moderate negative relationship with spending score.
- Income and spending score have almost no linear correlation.
- Income and spending patterns reveal distinct customer groups suitable for clustering.

CONCLUSION AND LIMITATIONS

- K-Means and Hierarchical Clustering can effectively identify customer groups based on similarities in their characteristics and purchasing behavior.
- Customer segmentation helps businesses understand their customers and create personalized marketing campaigns, targeted offers, and improved customer experiences

LIMITATIONS:

- Results depend on the quality of the dataset.
- Choosing the number of clusters can affect K-Means results.
- Customer behavior may change over time.
- Clustering results may not always have clear business interpretations.
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